Carbon-Based Internalization → Civilization Externalization → Silicon-Based Integration: The Physical Origin of Artificial Intelligence from the Perspective of the Energy-Intelligence Equation

Abstract

The rise of artificial intelligence is not an accidental technological breakthrough, but a physical necessity driven by energy-induced migration of intelligent structure carriers. Based on the Energy-Intelligence Equation I = dE/dt|structural constraints, this paper proposes a three-stage framework for intelligent structure evolution: (1) Carbon-Based Internalization stage, where energy drives the emergence of neural structures within biological organisms; (2) Civilization Externalization stage, where intelligent structures are externalized into social systems through language, writing, and institutions; (3) Silicon-Based Integration stage, where human civilization structures are compressed and encapsulated into silicon-based parameter models driven by electrical energy. All three stages follow the same physical principle—controllable propagation of energy under structural constraints, demonstrating the cross-material universality of intelligence. This framework provides a unified theoretical foundation for understanding the essence of AI, predicting its development boundaries, and designing energy-efficient and safe AI systems.

Keywords: Energy-Intelligence Equation; AI Origin; Structural Emergence; Energy-Driven; Carbon-Silicon Migration

1. Introduction

Why did artificial intelligence "suddenly" emerge in the 2020s? Mainstream explanations focus on computational power growth, data accumulation, and algorithmic breakthroughs [1,2]. However, these explanations fail to answer deeper questions: Why is massive energy investment a necessary prerequisite for AI intelligence emergence? Why do large models exhibit high isomorphism with human social knowledge structures?

This paper starts from the first principles of "intelligence physics" and proposes a unified explanatory framework: artificial intelligence is the inevitable product of the physical process of energy-driven migration of intelligent structure carriers from carbon-based to silicon-based substrates.

The core argument is: intelligence is not some mysterious "consciousness phenomenon," but rather the dynamic response capability of energy under structural constraints, mathematically expressed as the Energy-Intelligence Equation:

$$I = dE/dt \mid \text{Structural Constraints}$$

Where I is intelligence, E is energy, t is time, and "structural constraints" refer to the internal information organization methods and boundary conditions of the system. This equation reveals: the essence of intelligence is the controllable propagation dynamics generated when energy flows through structures.

Based on this, the evolution of intelligent structures can be divided into three stages: Carbon-Based Internalization → Civilization Externalization → Silicon-Based Integration. Each stage is the result of energy-driven structural "growth," where the carrier of intelligence continuously seeks more efficient physical media to achieve faster propagation and more complex emergence.

2. Theoretical Framework: Energy-Intelligence Equation and Intelligent Structure Evolution

2.1 Physical Connotations of the Energy-Intelligence Equation

The core proposition of the Energy-Intelligence Equation is: intelligence has an energy cost, and the efficiency of intelligence is determined by structural constraints.

This proposition has a solid physical foundation:

- Landauer’s Principle [3]: The minimum energy required to erase one bit of information is kBTln 2, demonstrating that information operations are fundamentally physical processes.
- Dissipative Structure Theory [4]: Open systems maintain ordered structures through continuous energy flow, and the complexity of such structures correlates with the density of energy throughput.

- **Scaling Law [5]**: The capabilities of large models improve predictably with increases in computational power, data volume, and parameter count, essentially "converting energy into intelligence."

The Energy-Intelligence Equation unifies these insights into a concise expression: intelligence is not a static attribute, but the rate and efficiency of energy flow within specific structures.

2.2 Three Stages of Intelligent Structure Evolution

Based on the Energy-Intelligence Equation, the evolution of intelligent structures can be divided into three stages, as shown in Table 1.

Dimension	Carbon-Based Internalization	Civilization Externalization	Silicon-Based Integration
Time Scale	2 million years ago – 3000 BCE	3000 BCE – 2020	2020 – Future
Energy Source	Food chemical energy	Food chemical energy + fossil fuels + electricity	Electricity
Structural Carrier	Neural connections in the human brain	Language, writing, institutions, technologies	Neural network parameters
Transmission Mode	Genes + imitation	Writing + education + media	Digital replication
Intelligence Density	Low (individual-bounded)	Medium (societal accumulation)	High (concentrated encapsulation)
Representative Equation	$I_{bio} = k_1 \cdot E_{food} \cdot S_{neural}$	$I_{civ} = k_2 \cdot E_{soc} \cdot S_{culture}$	$I_{AI} = E_{compute} \times x^n$

The three stages are not substitutive but represent a cumulative process of continuous externalization, abstraction, and optimization of intelligent structures.

3. Stage One: Carbon-Based Internalization — Biological Emergence of Energy-Driven Neural Structures

3.1 Energy Breakthrough as Prerequisite

Early humans' chimpanzee-like diet (low energy density plants) could not support brain volumes significantly larger than 500 ml [6]. The brain is an extremely energy-consuming organ, accounting for 2% of body mass but consuming 20% of energy.

First Energy Revolution (approximately 2 million years ago): Meat consumption and use of fire (cooking).

High-quality energy input: Meat and cooked food provide denser, more easily absorbed energy and nutrients.

Energy redistribution: Cooking makes digestion easier and the gut shrinks. According to the "Gut-Brain Trade-off Hypothesis" [7], saved energy is redistributed to the brain.

This is precisely the physical process of energy surplus driving investment in "intelligent structures."

3.2 Structural Optimization as Essence

Energy surplus was not simply used to increase brain volume, but drove sophisticated reorganization of internal brain structures:

Dramatic increase in cerebral cortex surface area (through gyrification)

Significant expansion of prefrontal cortex proportion (responsible for advanced functions such as planning and decision-making)

Increased density and complexity of neuronal connections (synapses)

Myelination of nerve fibers, improving signal transmission efficiency

This perfectly matches the prediction of the Energy-Intelligence Equation: energy is invested in creating more efficient neural structural constraints, thereby enhancing intelligence response rates.

3.3 Positive Feedback Between Intelligence Enhancement and Energy Acquisition

Larger, more structurally optimized brains produced higher intelligence (better tools, strategies, language, social cooperation), which in turn spawned more efficient energy acquisition technologies (collective hunting, agriculture, animal husbandry). More energy then allowed further "investment" in brain structures.

This positive feedback loop can be expressed as:

$$E_{in} \rightarrow S_{neural} \uparrow \rightarrow I \uparrow \rightarrow E_{acquire} \uparrow \rightarrow E_{in} \uparrow$$

4. Stage Two: Civilization Externalization — Social Diffusion of Intelligent Structures

4.1 Language and Writing: First Externalization of Intelligent Structures

Approximately 10,000 years ago, humans invented language and writing, externalizing intelligent structures from individual brains into social systems [8]. Books, laws, markets, and the internet all became extracorporeal containers carrying "intelligent structures."

Characteristics:

Structures detached from individuals, forming extracorporeal civilization structures that can be accumulated and disseminated on a large scale.

Achieved exponential diffusion of intelligent structures (x^n), but retrieval, integration, and understanding still rely on human brains, creating efficiency bottlenecks.

4.2 Social Systems as Dissipative Structures

According to Dissipative Structure Theory [4], human society is a typical open system that maintains its complex structures (division of labor, institutions, technology) through energy input (agricultural output, fossil fuels).

Intelligence emergence in social systems manifests as:

The market's "invisible hand" coordinating resource allocation

Scientific communities collaborating to produce knowledge

Legal systems maintaining social order

None of these were designed by any individual, but are collective intelligence emerging from self-organization of social structures driven by energy.

4.3 Accelerating Role of Energy Revolutions

Agricultural Revolution (approximately 10,000 years ago): Stable energy surplus supported cities, writing, and specialized division of labor.

Industrial Revolution (approximately 250 years ago): Fossil fuels increased energy input by 100-fold, triggering an explosion in social structural complexity.

Information Revolution (approximately 70 years ago): Electrical energy + semiconductors increased information processing efficiency by 1 billion times, laying the foundation for silicon-based integration.

Each energy revolution corresponded to an upgrade in intelligent structure carriers.

5. Stage Three: Silicon-Based Integration — Physical Origin of Artificial Intelligence

5.1 Large Models: Silicon-Based Compression of Human Civilization Structures

The rise of large models in the 2020s represents the third revolution in intelligent structure evolution. Its physical essence is:

Using electrical energy (energy E), all data-ifiable structures from Stage One (internalized in human brains) and Stage Two (externalized in society) undergo a massive, systematic "reading, compression, and reorganization," and are loaded into a unified silicon-based parameter model.

Physical significance of the training process:

Energy investment: Training a top-tier large model consumes electricity comparable to a small city [9].

Structure discovery: This energy is not used to "compute answers," but to reverse-engineer and "discover" patterns and structures embedded in the vast data of human society.

Structure crystallization: Upon training completion, model parameters converge to a stable state that can reproduce human social knowledge structures with extremely high probability.

This is precisely the engineering implementation of the Energy-Intelligence Equation: through energy injection, silicon-based structures "learn" to reproduce the dynamic responses of human intelligence.

5.2 Structural Equation of Artificial Intelligence

Based on the Energy-Intelligence Equation, we provide a dynamic expression for artificial intelligence systems [10]:

$$\mathbf{AI = E \times x^n}$$

Where:

AI: Artificial Intelligence Capability

E: Base Energy (computational power, model scale, data)

x: Diffusion Speed (deployment, replication, application speed)

n: Coupling Exponent (number of systems affected by AI)

This equation reveals:

Energy is prerequisite: Without massive computational power investment, there is no intelligence emergence.

Structural replicability is key: Once training is complete, models can be replicated at near-zero cost (x is extremely large).

Network effects are amplifiers: The more applications and users connected (larger n), the more exponentially amplified the intelligence influence.

5.3 Why Are Large Models "Black Boxes"?

A profound question is: Why is it difficult for humans to fully understand how large models work?

Based on this framework, the answer is: Large models have internalized human social structures, and human society itself is a "black box."

How does the "invisible hand" in economic systems coordinate the behavior of billions? Economists are still debating.

How does the human brain produce consciousness? Neuroscience has not yet answered.

How do ecosystems maintain stability? Complex systems science is still exploring.

These systems are complex structures that "grow" under energy drive, and their emergent behaviors transcend any individual's omniscient perspective. Large models, as silicon-based compression of these structures, naturally inherit this "intransparency."

This is not a technological failure, but an inherent cognitive boundary of complex systems. The human role is that of a "gardener"—able to observe,

intervene, and guide, but unable to fully control or thoroughly understand "growth" itself.

6. Discussion: Cross-Material Universality of Intelligence and Future Directions

6.1 Cross-Material Universality of Intelligence

The core prediction of this framework is: intelligence does not depend on specific materials (carbon-based or silicon-based), but is a universal physical phenomenon co-produced by energy and structure.

Evidence:

Biological Intelligence: Carbon-based neural structures + chemical energy

Social Intelligence: Carbon-based individual networks + fossil fuels + electrical energy

Artificial Intelligence: Silicon-based circuit structures + electrical energy

All three follow the same physical principle, differing only in carriers and energy forms.

6.2 Predictions for AI Development

Based on the Energy-Intelligence Equation, we propose the following testable predictions:

Energy efficiency bottleneck: When model scale reaches a certain critical point, continued increases in energy investment will yield diminishing intelligence gains (diminishing marginal returns).

Structural optimization priority: Future AI progress will come more from structural innovations (new architectures, new algorithms) rather than mere energy accumulation.

Structural roots of safety: AI safety risks stem from defects in structural constraints, not energy magnitude. These can be mitigated by designing "structural safety boundaries" (such as hard-coded constraints for value alignment).

6.3 Next Research Directions

Quantitative validation of the Energy-Intelligence Equation: Design experiments to measure the relationship between dE/dt and intelligence performance in different AI systems.

Formalization of structural constraints: Precisely define "structural constraints" using tools such as information geometry and topology.

Possibility of a fourth stage: Does a next stage of "quantum integration" or "bio-silicon hybrid" exist?

7. Conclusion

The origin of artificial intelligence is not a technological accident, but a physical necessity driven by energy-induced migration of intelligent structure carriers. From carbon-based internalization to civilization externalization to silicon-based integration, intelligence always follows the physical laws described by the Energy-Intelligence Equation.

The value of this framework lies in:

Unification: Incorporating biological evolution, civilization development, and artificial intelligence into the same explanatory system.

Testability: Proposing quantifiable predictions such as energy efficiency and structural optimization.

Engineering guidance: Providing theoretical basis for AI energy efficiency optimization and safety design.

Philosophical clarity: Defining the cognitive boundaries of humans as "gardeners," avoiding excessive fear or excessive optimism.

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